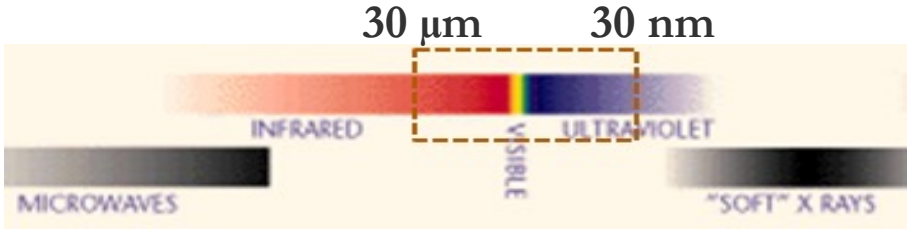
Overview Lecture

Photonics

Photonics is the science and engineering of the harnessing of light.



Involves, the Generation of light

the Detection of light

the Management of light
(guiding, manipulation and amplification)

the Utilization of light
as a tool for the benefit of mankind

Photonics

Electrical Engineering:	The Electromagnetic wave nature of light. Waveguiding and Mode Theory. Electronics and Optoelectronics. Communication Theory.
Physics/Chemistry:	Solid-state physics and Quantum mechanics.
Materials Science and Engineering:	Structure and Characterization. Crystallography. EOM and other properties. Synthesis and processing. Nanomaterials.
Others (on demand):	*for example*, Biomedical.

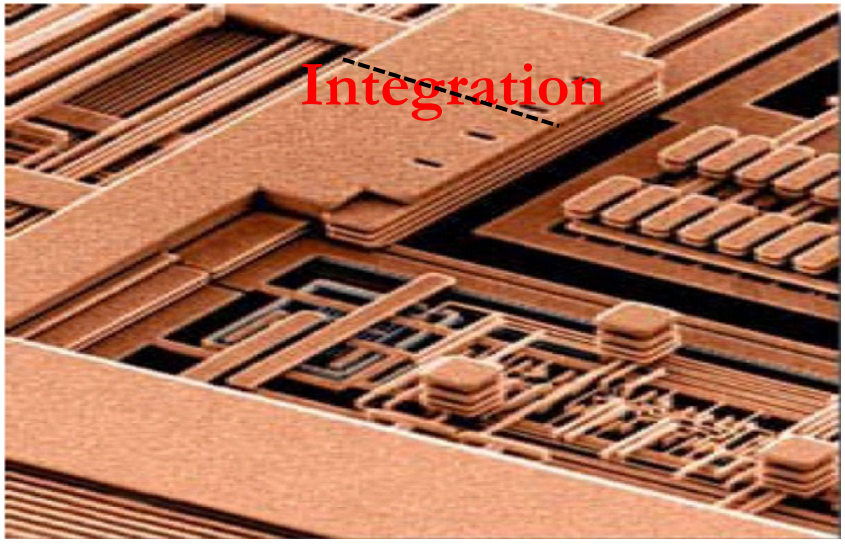
Photonics Vacuum Tube Era

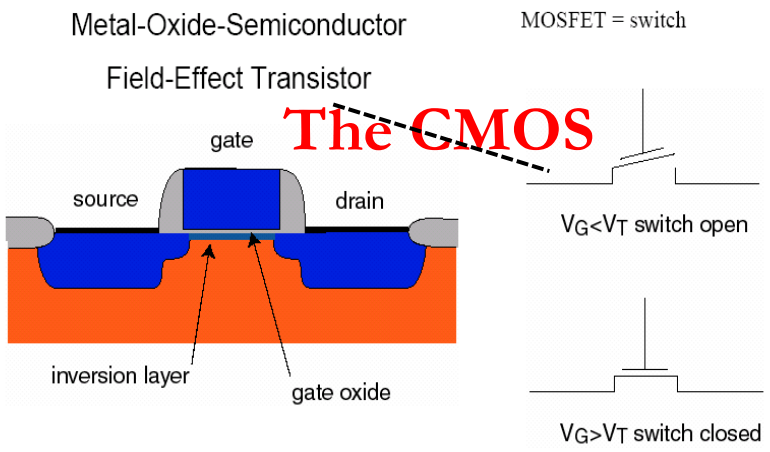
Outstanding Issues

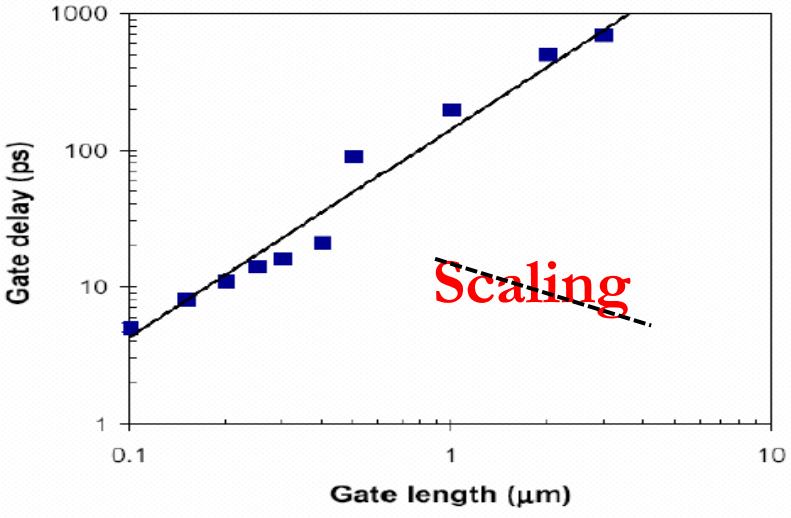
- ✓ Issues related to the nature of light.
- ✓ Issues related to light-matter interaction.
- ✓ Issues related to engineering design.
- ✓ Issues related to economic feasibility.

Photonics vs. Electronics

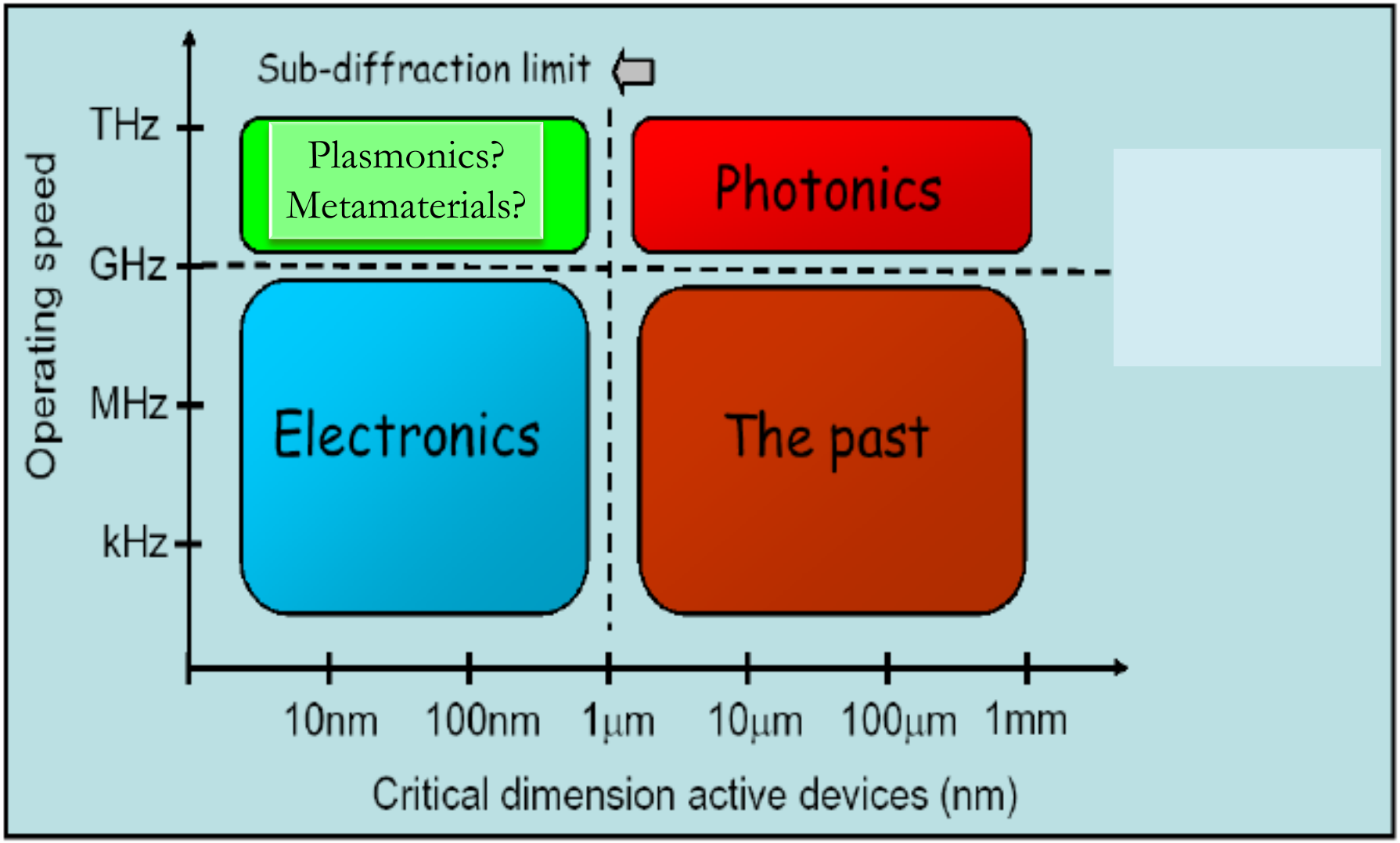








Meeting the Challenges

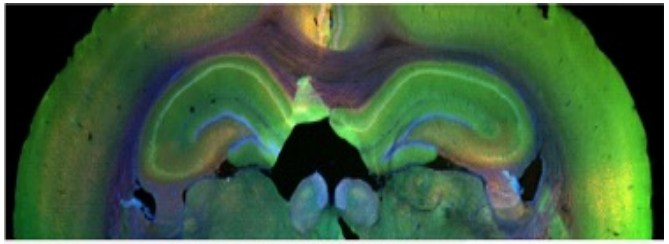


Photonics Potential

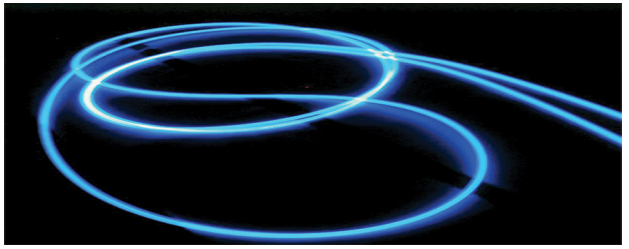
Nano Photonics and Plasmonics Research



UV water purification system



Fluorescence microscopy of stained section of rat brain





Solar lighting



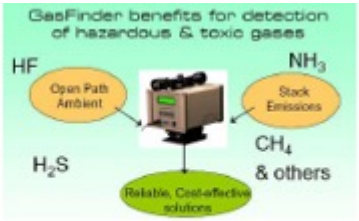
Near IR spectroscopy of skin lesions

Spartan



Rapid DNA analysis





Detection of hazardous gases and pollutants





Infrared night sights



Enablence GPON transceivers



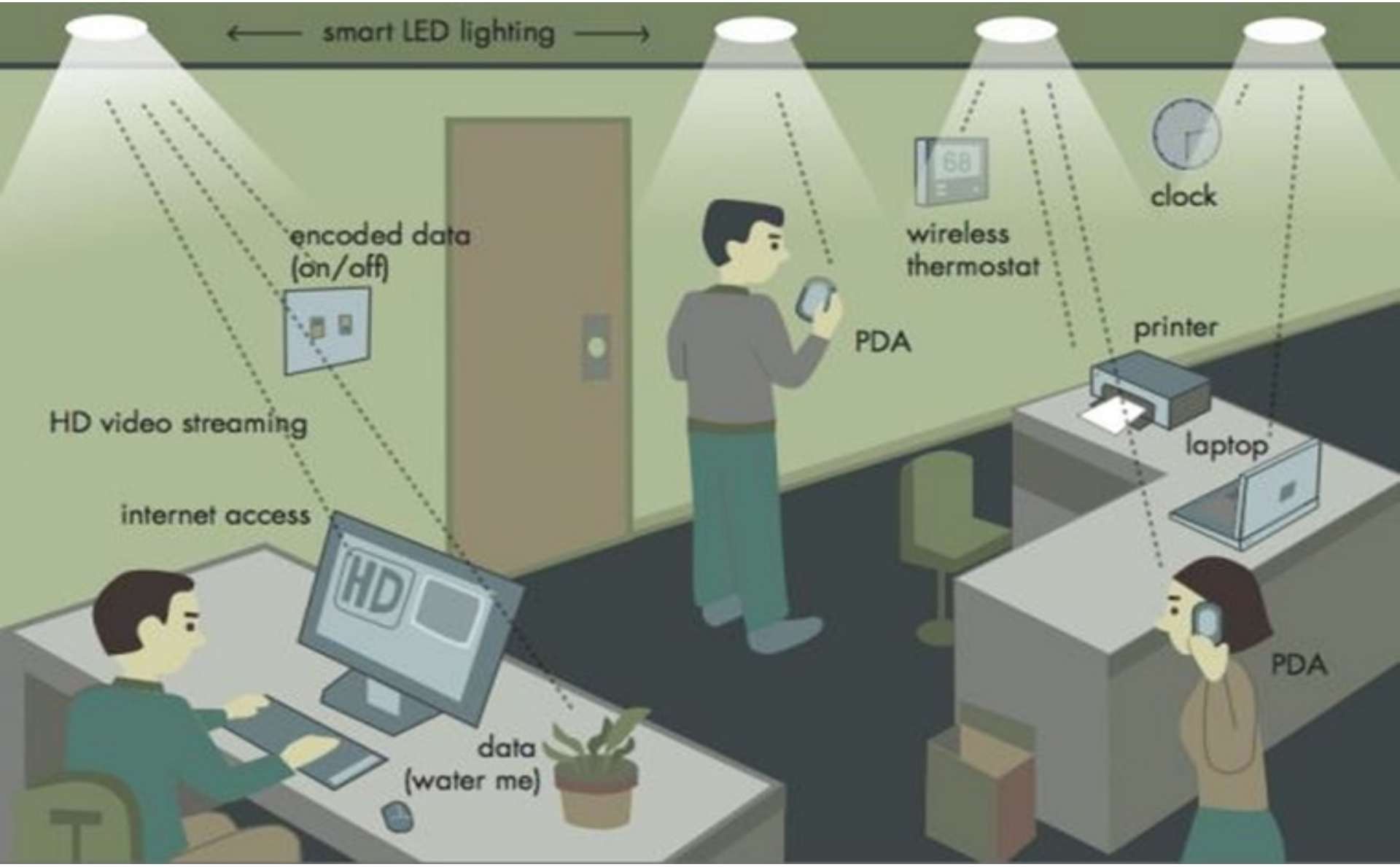
Optech's *LYNX* Mobile Mapper

Major Photonics Fields

Photonics will take over Electronics and other technologies in many important fields.

- ✓ Optical Computing.
- ✓ Optical Communications.
- ✓ Energy.
- ✓ Biomedical applications.
- ✓ Solid-state lighting.

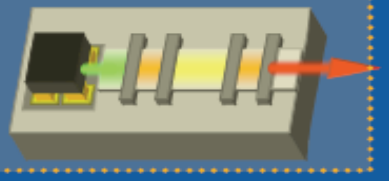
Solid-state Lighting



The Future is Here!

Intel's Silicon Photonics Research

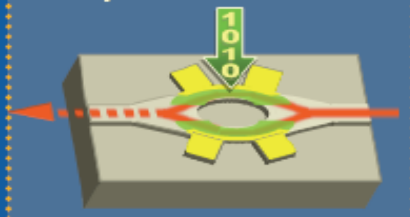
1) Light Source



2) Guide Light

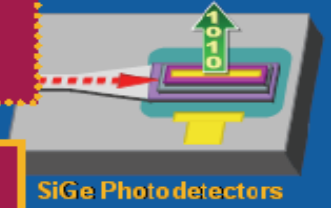


3) Modulation

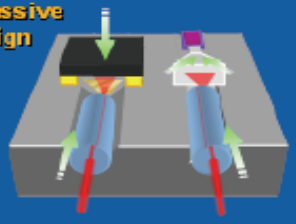


Continuous Wave Silicon Raman Laser *(Feb '05)*

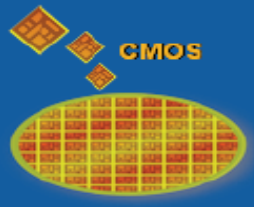
4) Photo-detection



5) Low Cost Assembly



6) Intelligence



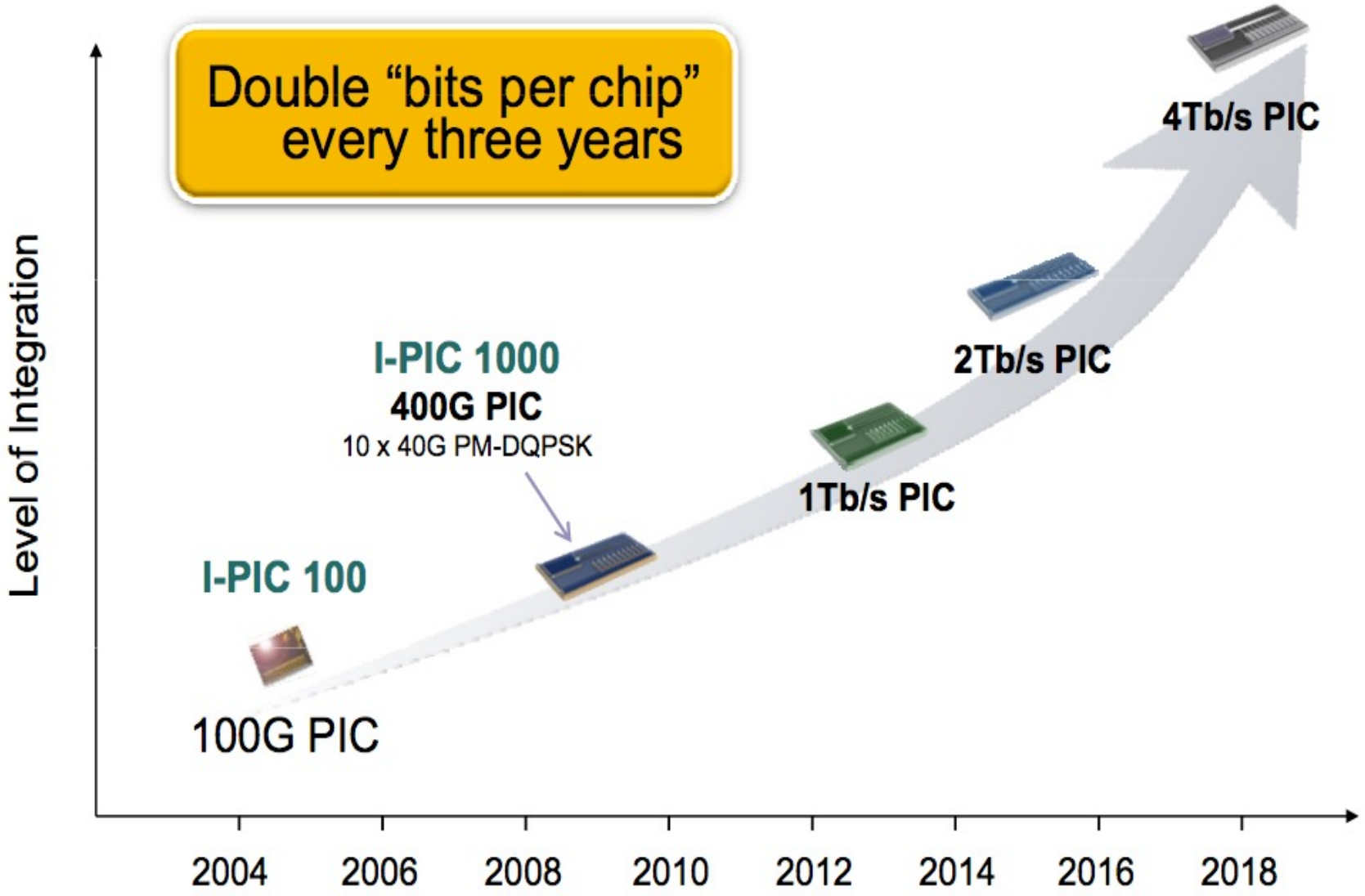
1GHz *(Feb '04)*
10 Gb/s *(Apr '05)*

40 Gb/s (Jul '07)

Electrically Pumped Hybrid Silicon laser
(September 2006)

We have achieved 40 Gb/s milestone
Focus is now on integration

The Future is Here!



Course Topics

What is light?

How do we generate light?
Man-made light

How do we detect light?

How do we manage light?

How do we use light?

Syllabus

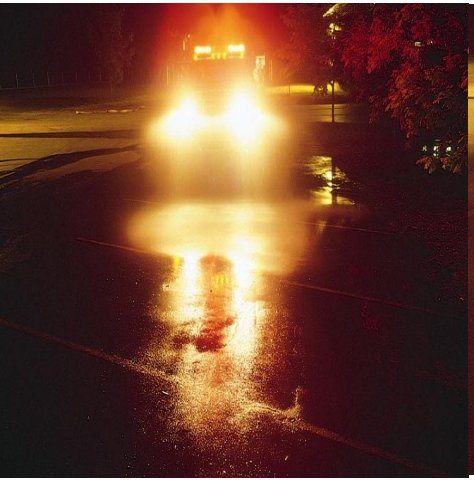
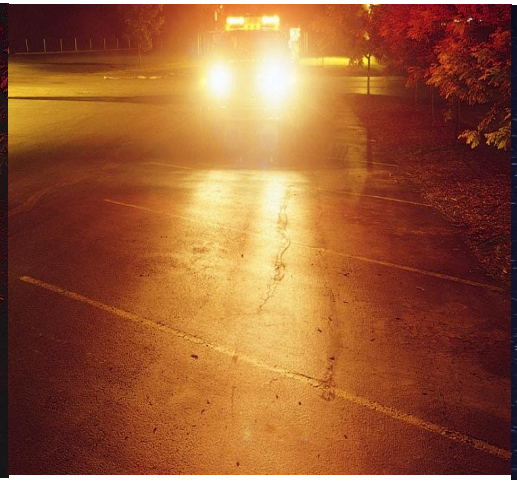
Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Office Hours MW: 11:30-12:30 T: 13:00-14:00	Phone ---	Email mohammad.alsunaidi@marmara	
Course Information	Text: Saleh/Teich, *Fundamentals of Photonics*	Grading	Class Work Assignments (3x5)% + Project 10%		Midterm Exam 35%	Final Exam 40%

Week	Events	Subjects	Mondays	Tuesdays
1 Feb 27-Mar 3	Course drop/add Week	The Theory of Light	Course introduction: The nature of light. Theory of light. Ray optics.	Ray optics: Prisms and Lenses
2 Mar 6-10	---		Ray optics: Lenses and Mirrors	Wave optics: Interference
3 Mar 13-17	---		Wave optics: Interference, Diffraction.	Wave optics: Diffraction.
4 Mar 20-24	Assignment 1 Due Mar 24		EM Wave optics: Maxwell's and Wave Equations. Types of waves.	EM Wave optics: Propagation.
5 Mar 27-31	---		EM Wave optics: Reflection. Refraction, Anti-reflection coating.	EM Wave optics: Polarization, Scattering.
6 Apr 1 – 9	**Mid-term Exams period**			
7 Apr 10-14	---		EM Wave optics: Light-matter interaction and the complex refractive index.	Quantum Optics. Photoelectric effect. The photon.
8 Apr 17-21	Holidays: Apr 20-23		Quantum Optics. Photoelectric effect. The photon.	Quantum Optics..
9 Apr 24-28	Labor Day: May 1	Optical Devices	Semiconductor fundamentals.	The PN Junction.
10 May 1-5	Midterm Make-up (Excused, May 2-7)		Photo-Detectors: Basics.	Photo-Detectors: pin and APD.
11 May 8-12	Assignment 2 Due May 12		Photo-Detectors: Characterization.	Photovoltaic Devices. The solar cell.
12 May 15-19	---		Light Emitting Devices (LEDs). Basics	Light Emitting Devices (LEDs). LED Structures and Characterization.
13 May 22-26	Assignment 3 Due May 12		LASERS. The Fabry-Perot Cavity.	Semiconductor Lasers.
14 May 29-June 2	Students Presentations	Special Topics	Students' Presentations.	Students' Presentations.
Jun 5 - 18	**Final Exams period**		**June 19-25 (Re-sit Final Exams period)**	

The Nature of Light



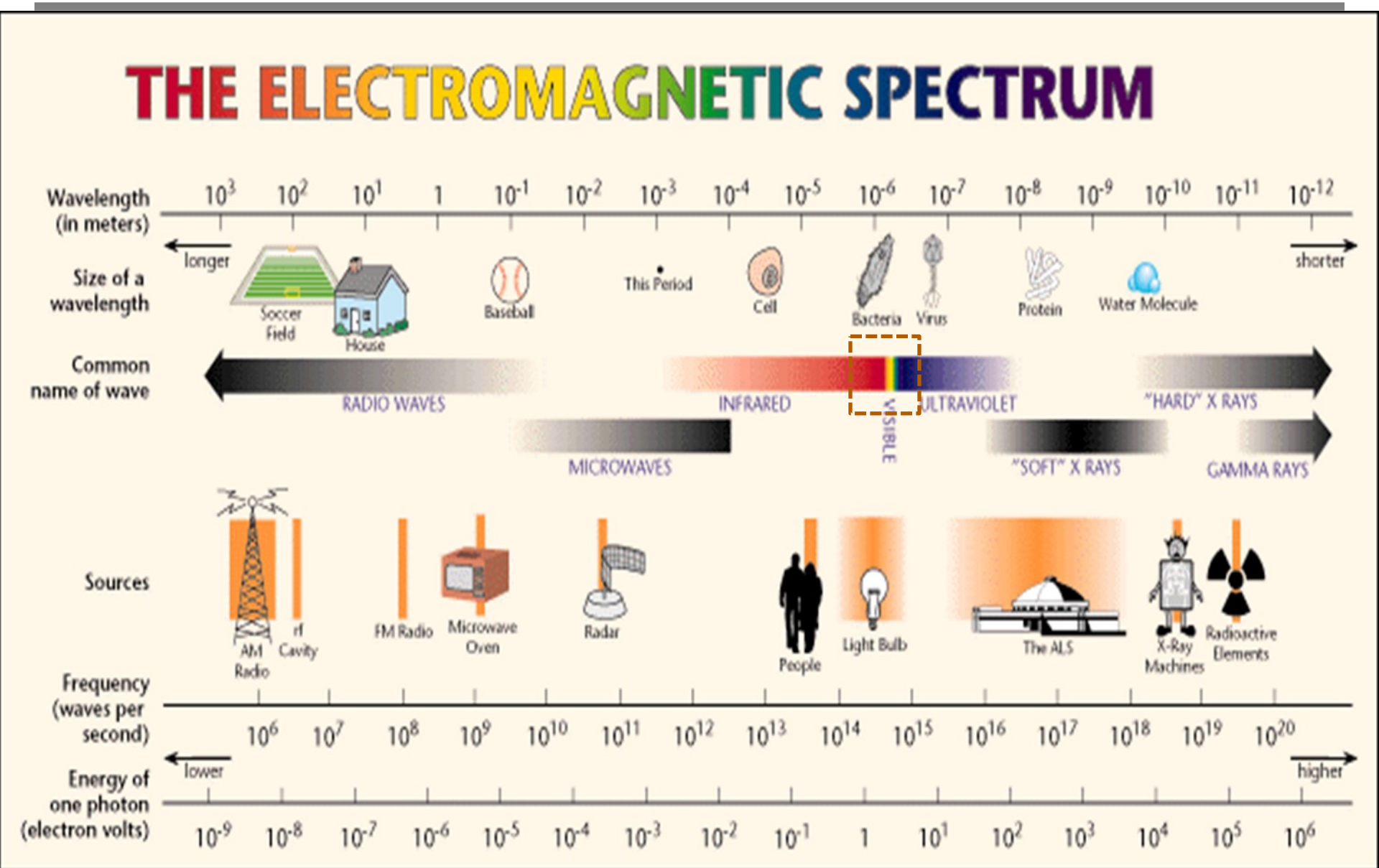




Light is waves! No, light is particles! No, No, Light is waves!
Wrong! light is particles!! Nope, Light is waves! Noooo, ….

The Optical Range



Natural and Man-made Sources

In general, Natural sources are free but produce incoherent, unpolarized and difficult to control light. Utilization is costly.
Chemi-luminescence Bio-luminescence.

Man-made sources can generate intense, focused, coherent, polarized and controllable light (Lasers). Controllable random light is also possible (Incandescent, LEDs). Some are compatible with electronics. They are cheap (but not free).

Others like Fluorescence and Phosphorescence.

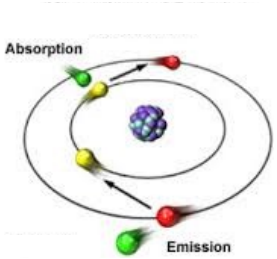
Light Generation Mechanisms

Charge Vibrations: Dipoles





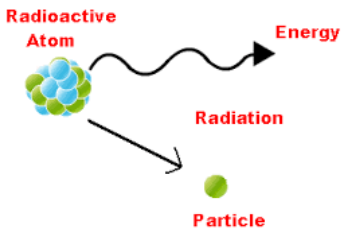

Electronic Transitions: EH Recombination







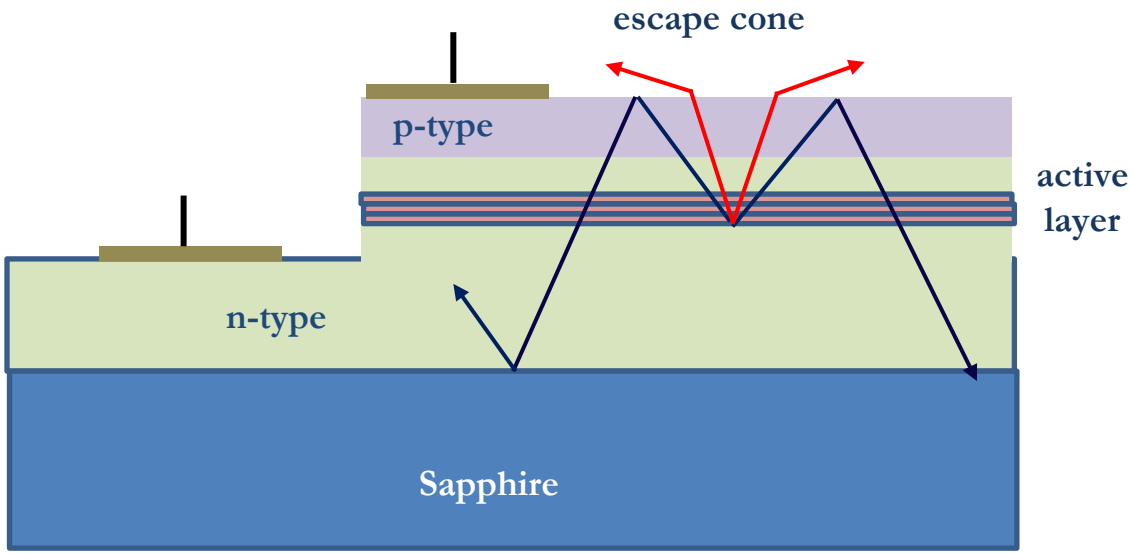
Atomic Disintegration: Decay






Light-emitting Diodes (LEDs)





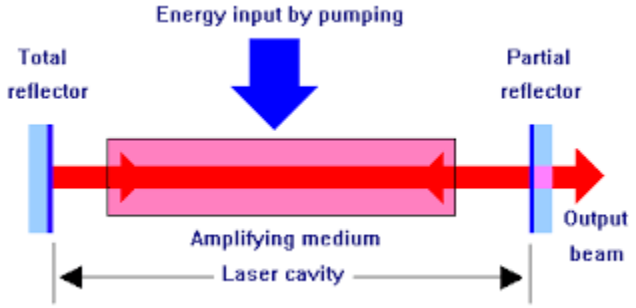
- Electron-hole recombination.
- Random unpolarized unfocused light.
- Low-energy light.
- Light color (wavelength) depends on material of active layer.
- Can produce white light by mixing blue, green and red.
- Only about 5% of light exits the device.
- Used to feed signal to optical fiber.

Optical Components: Sources

The LASER



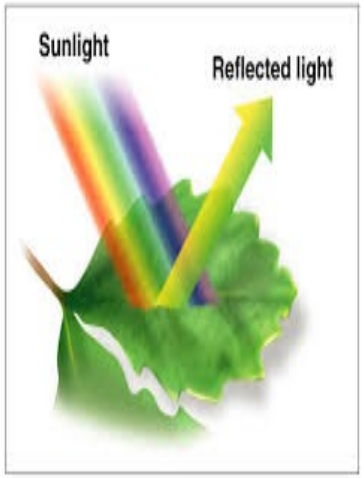
- Intense, polarized and focused light.



- Light color/wavelength depends on material.
- Only specific wavelength possible.
- Semiconductor lasers are low-energy compared to gas lasers.
- Used in various medical applications.

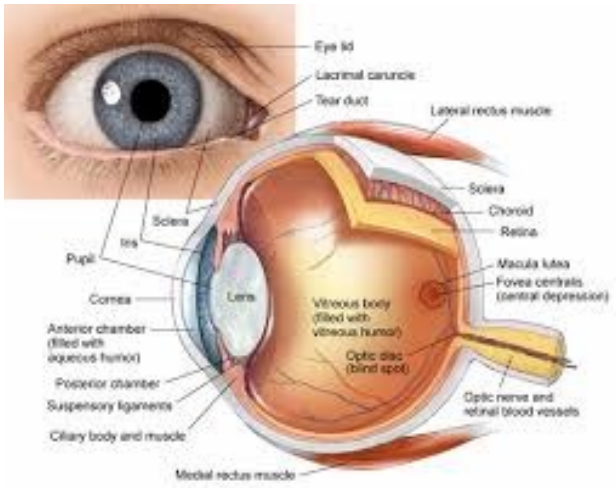
Light Detection Mechanisms

Natural Detectors

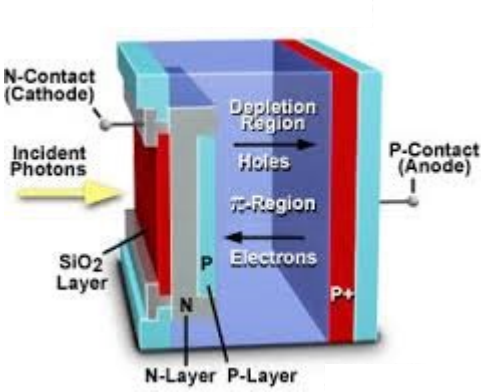








Man-made Detectors



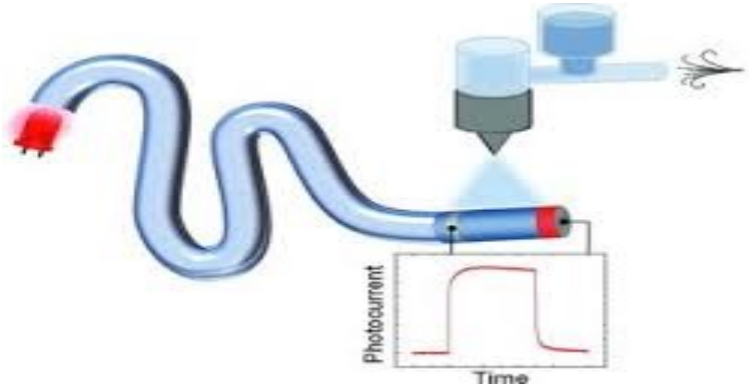




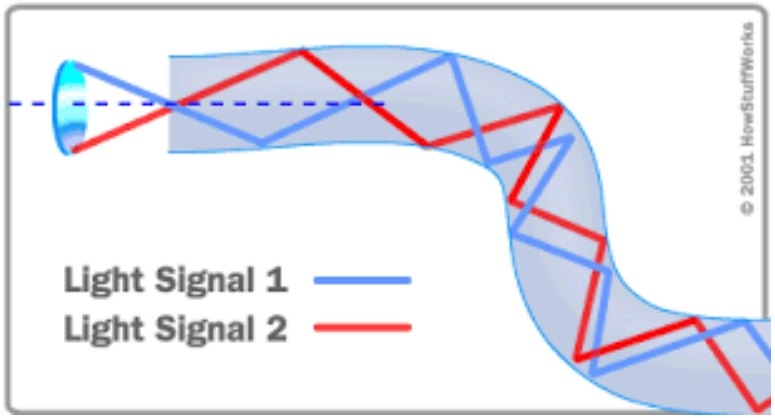
Optical Components: Waveguides

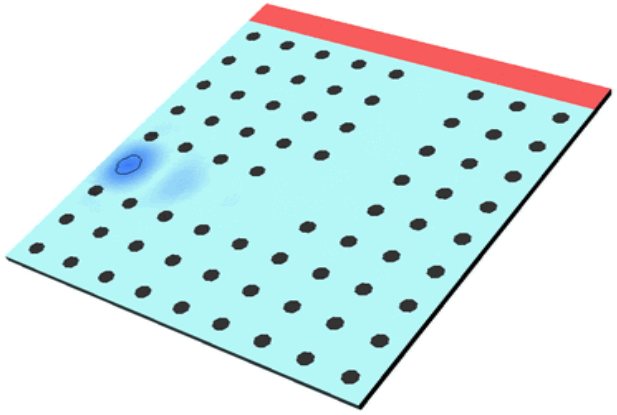


Discrete and Integrated



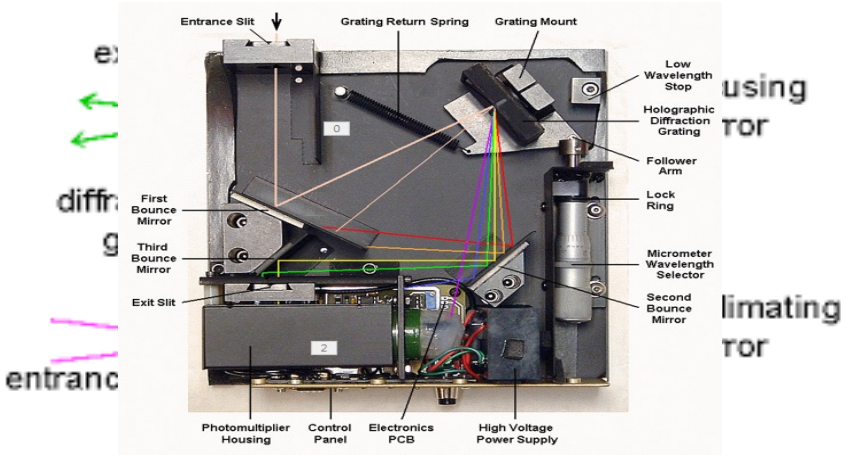
- Transfer multiple signals (without interference) to and from the target on the same channel.
- Speed and bandwidth are huge.
- Can have filters, amplifiers, etc. embedded.
- Loss can be controlled.





Optical Components: Others

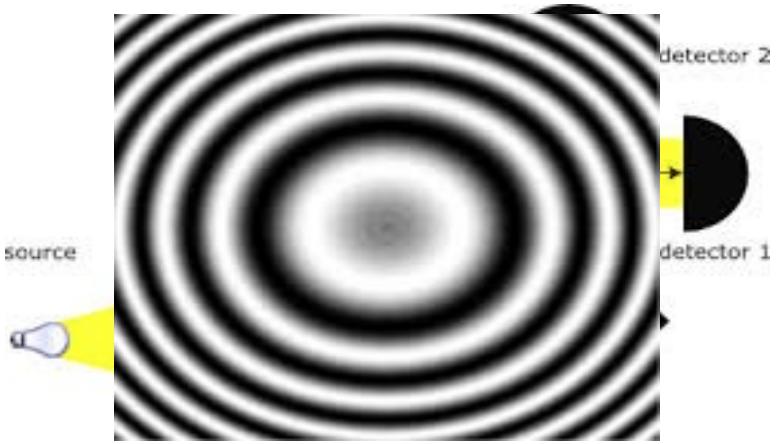
Monochromators



A beam of light containing many colors (wavelengths) can be split into beams with different colors (wavelengths) using a simple grating. This is particularly useful in producing scattering and absorption spectra.

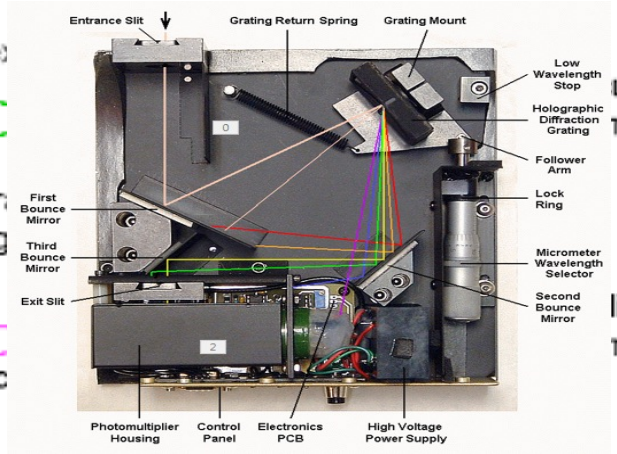
Interferometers

Two identical and in phase beams are produced using a 50-50 mirror (splitter). The beams travel different paths but eventually are merged at the output. Any change in phase due to path length, change in speed, etc. appears on the screen as interference patterns.



Optical Components: Others

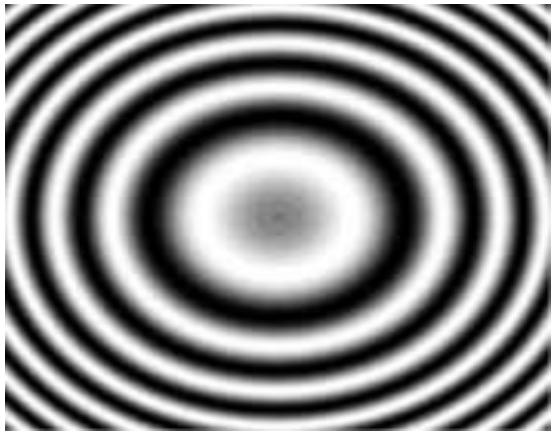
Monochromators



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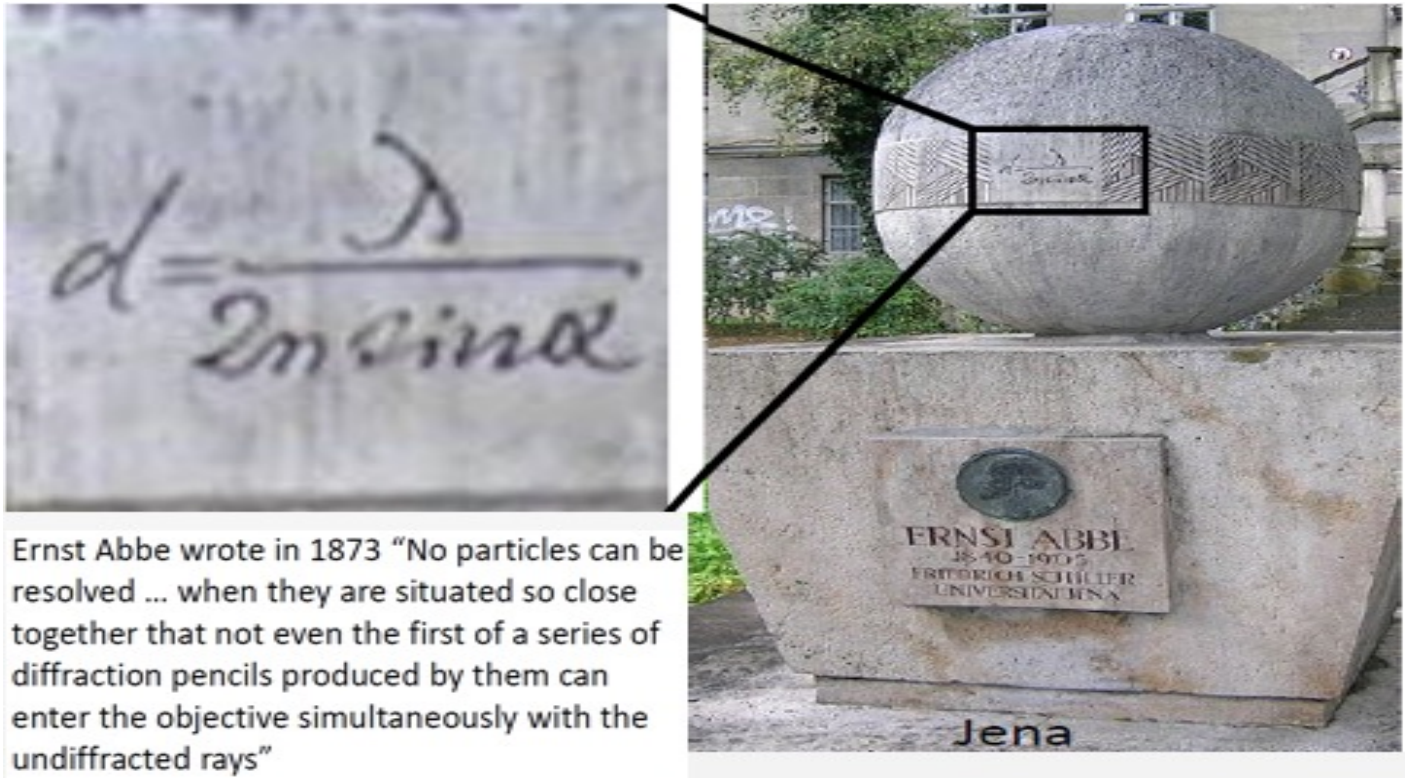
Interferometers

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Light Limitations

Detection by light waves is limited by the Diffraction Limit. Features less than half the wavelength in size cannot be resolved.



Ernst Abbe wrote in 1873 "No particles can be resolved ... when they are situated so close together that not even the first of a series of diffraction pencils produced by them can enter the objective simultaneously with the undiffracted rays"

Light Limitations

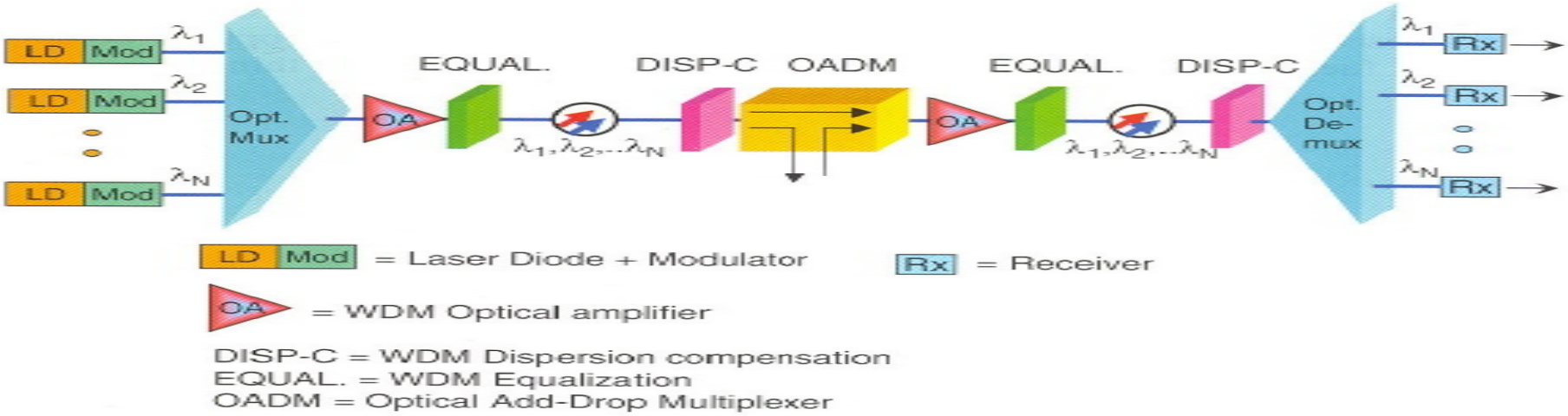
Detection by light waves is limited by the Diffraction Limit. Features less than half the wavelength in size cannot be resolved.

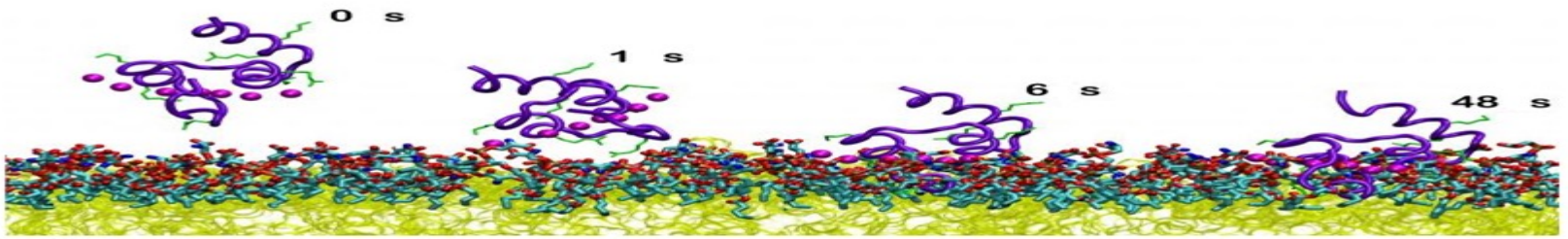
Light has enough energy to interfere with the material structure (electronic transitions, bonds, etc.).

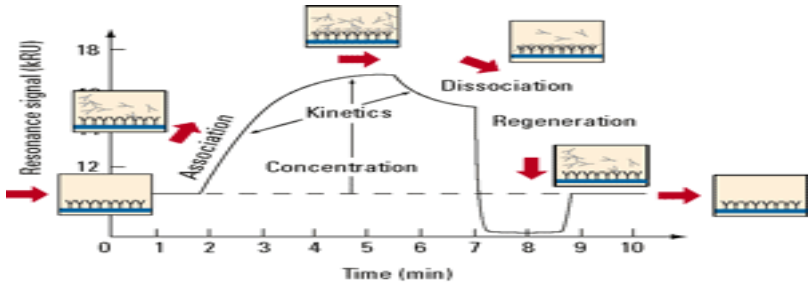
The spectrum of generated light is not continuous. Further material science research is needed to cover the gaps.

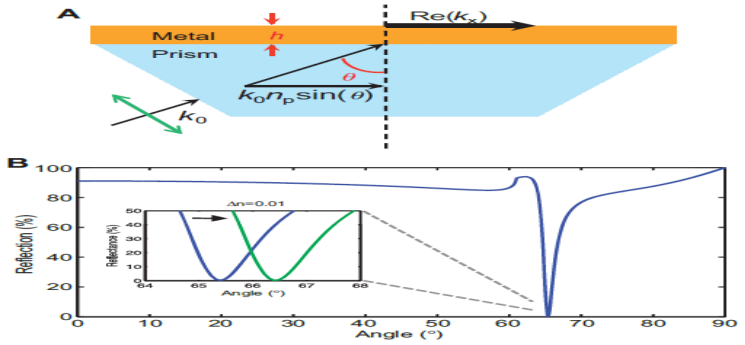
Light needs to be guided. Not easy!

Light Utilization/Applications









End of Part I

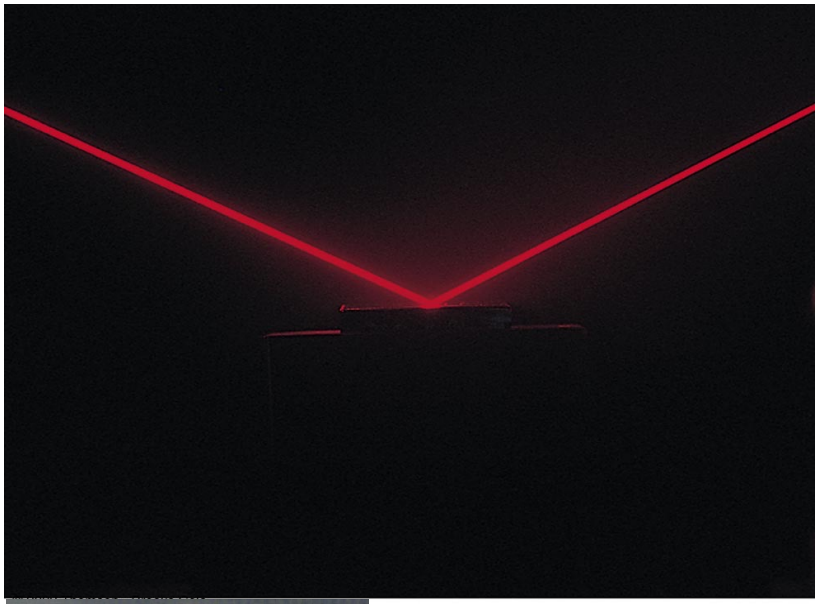
The Nature of Light

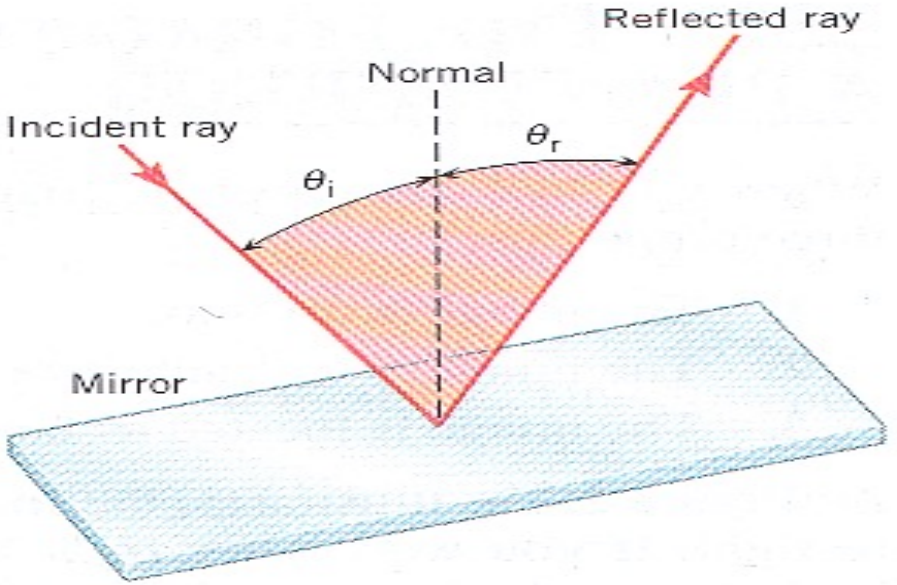
Greeks 600-200 BC: Particles from objects/from eye
Middle ages: Optical Engineering: light rays and beams
Modern age: Huygens: light behaves like waves!
Newton: No, light is particles
Young: Waves
Maxwell: Light is EM
Quantum mechanics: Mixed properties

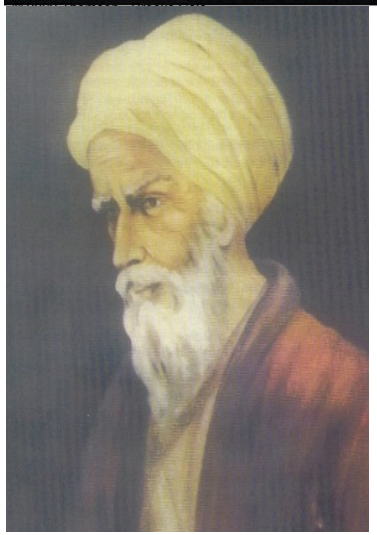
What is Light?

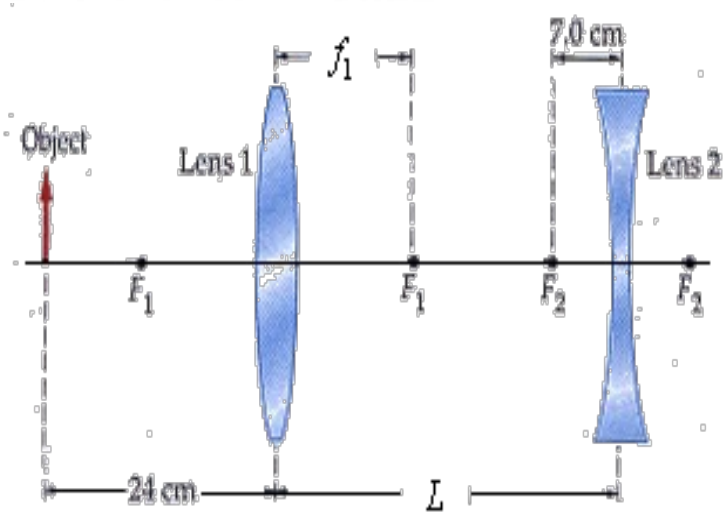


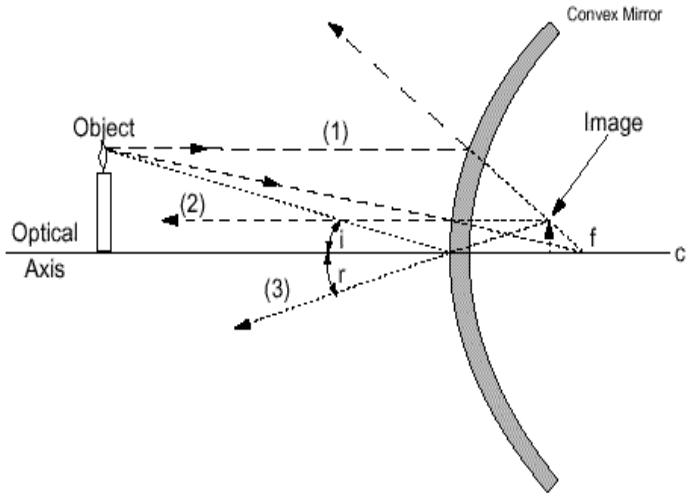
Optics




Reflected ray
Normal
Incident ray
θ_r
θ_i
Mirror

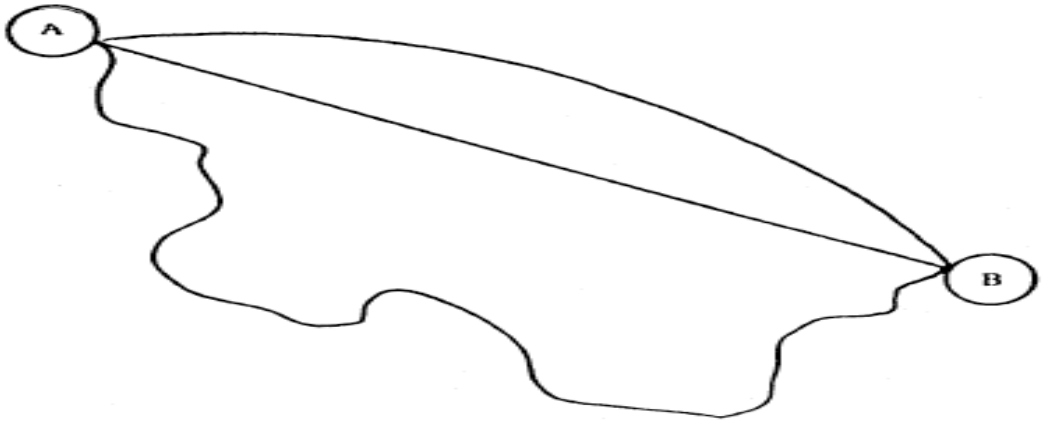


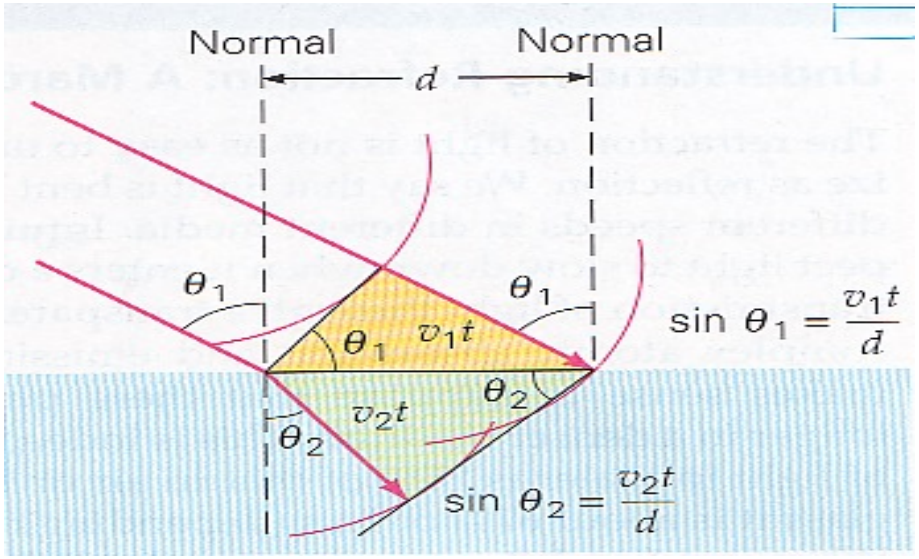

f_1
7.0 cm
Object
Lens 1
Lens 2
F_1
F_1
F_2
F_2
24 cm
L

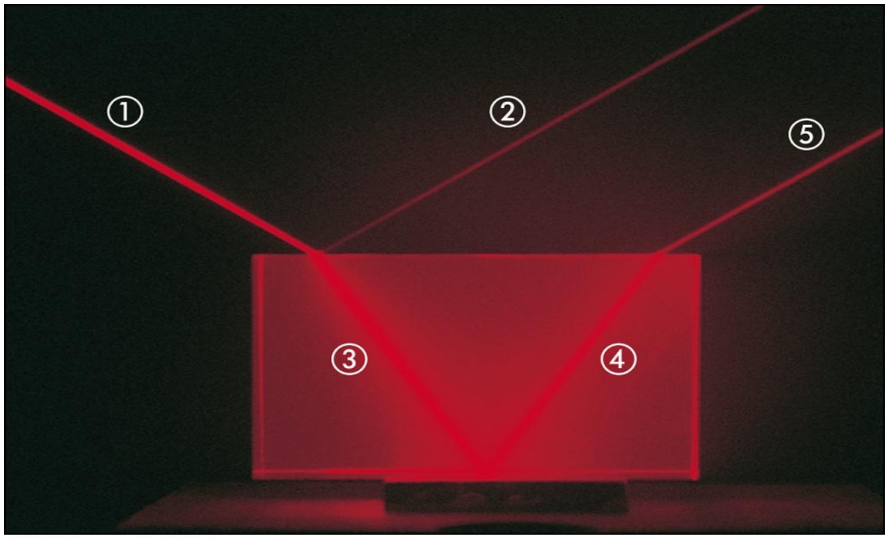

Convex Mirror
Object
(1)
Image
(2)
Optical
Axis
i
f
c
r
(3)

Snel's Law (1620) - Fermat's Law (1660)

A ray of light follows the path of "least time"







© 2003 Thomson - Brooks Cole

Snel's Work and Material Indexing (1620)

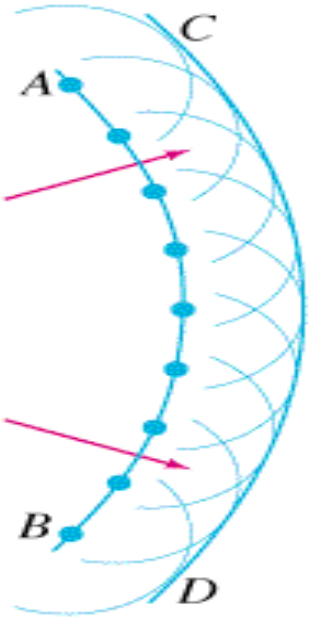
Substance	Refractive index
Air	1.00029
Water	1.33
Ethyl alcohol	1.36
Fused quartz	1.46
Glycerine	1.47
Glass	1.45-1.70
Oil	1.50
Zircon	1.92
Diamond	2.42

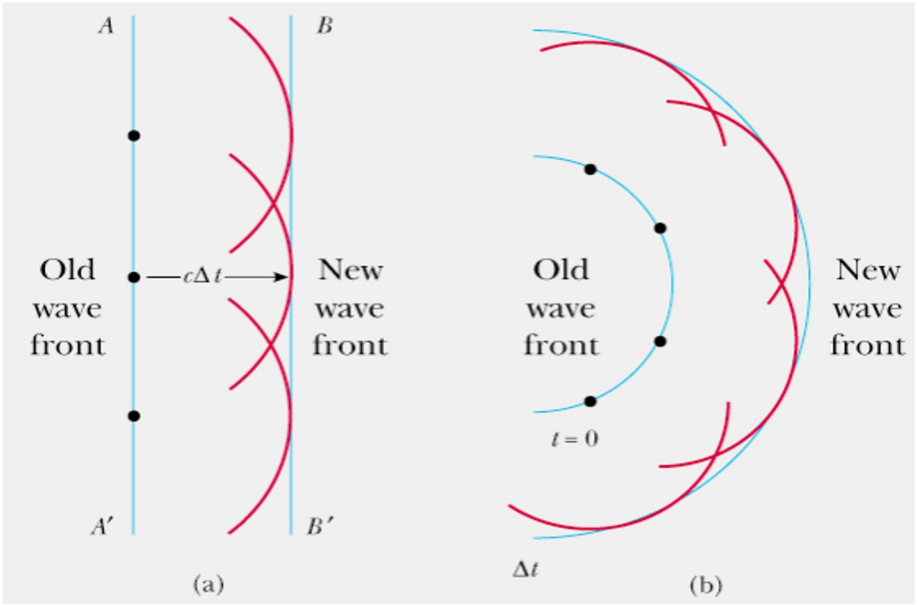
Huygens' Observations (1670)





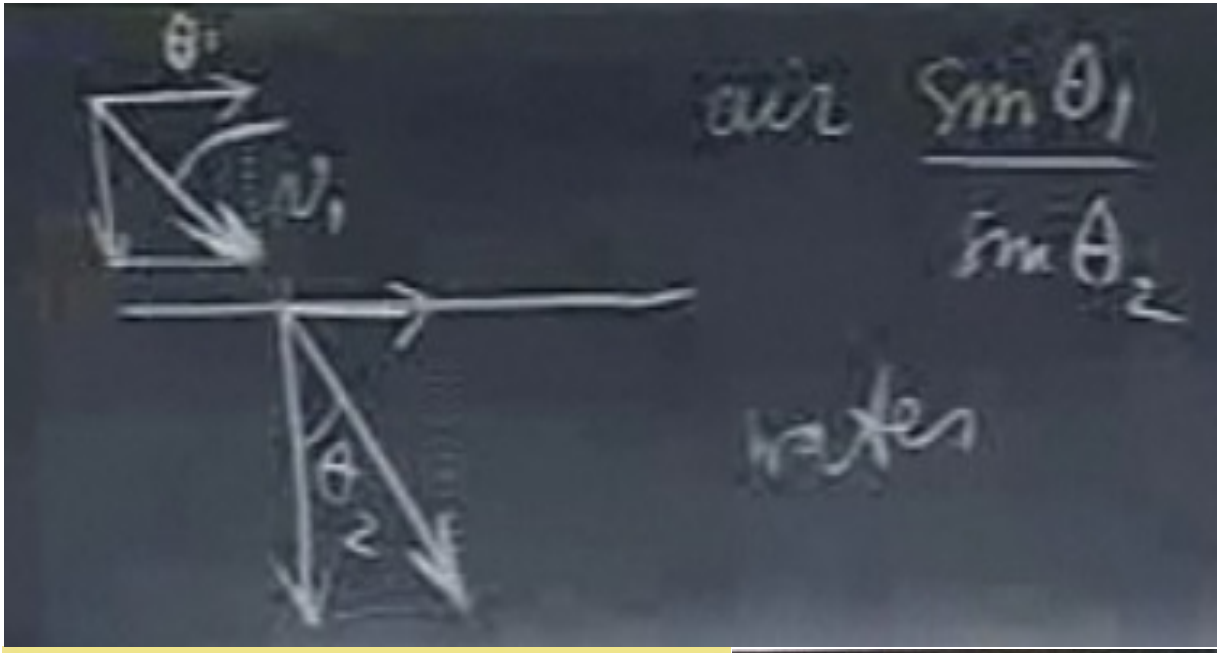






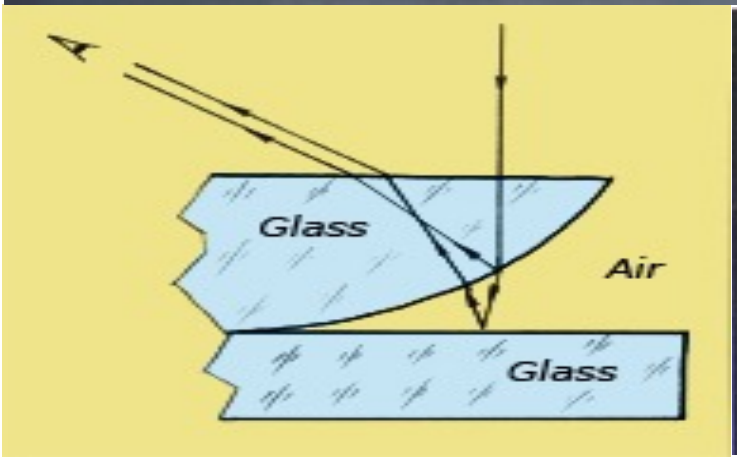
Newton's Theory (1700)

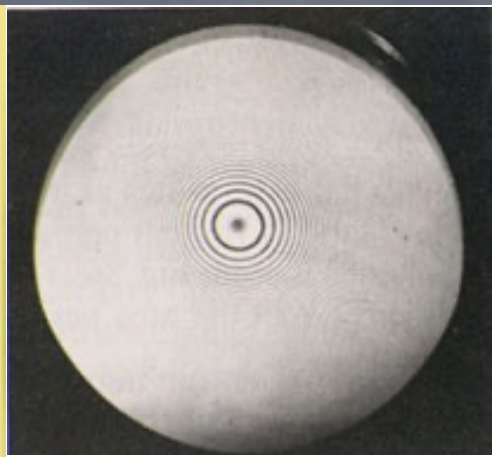
Are not the rays of light very small bodies emitted from shining objects?



$= v_2 / v_1$

Problem!!
Can we go over the speed of light

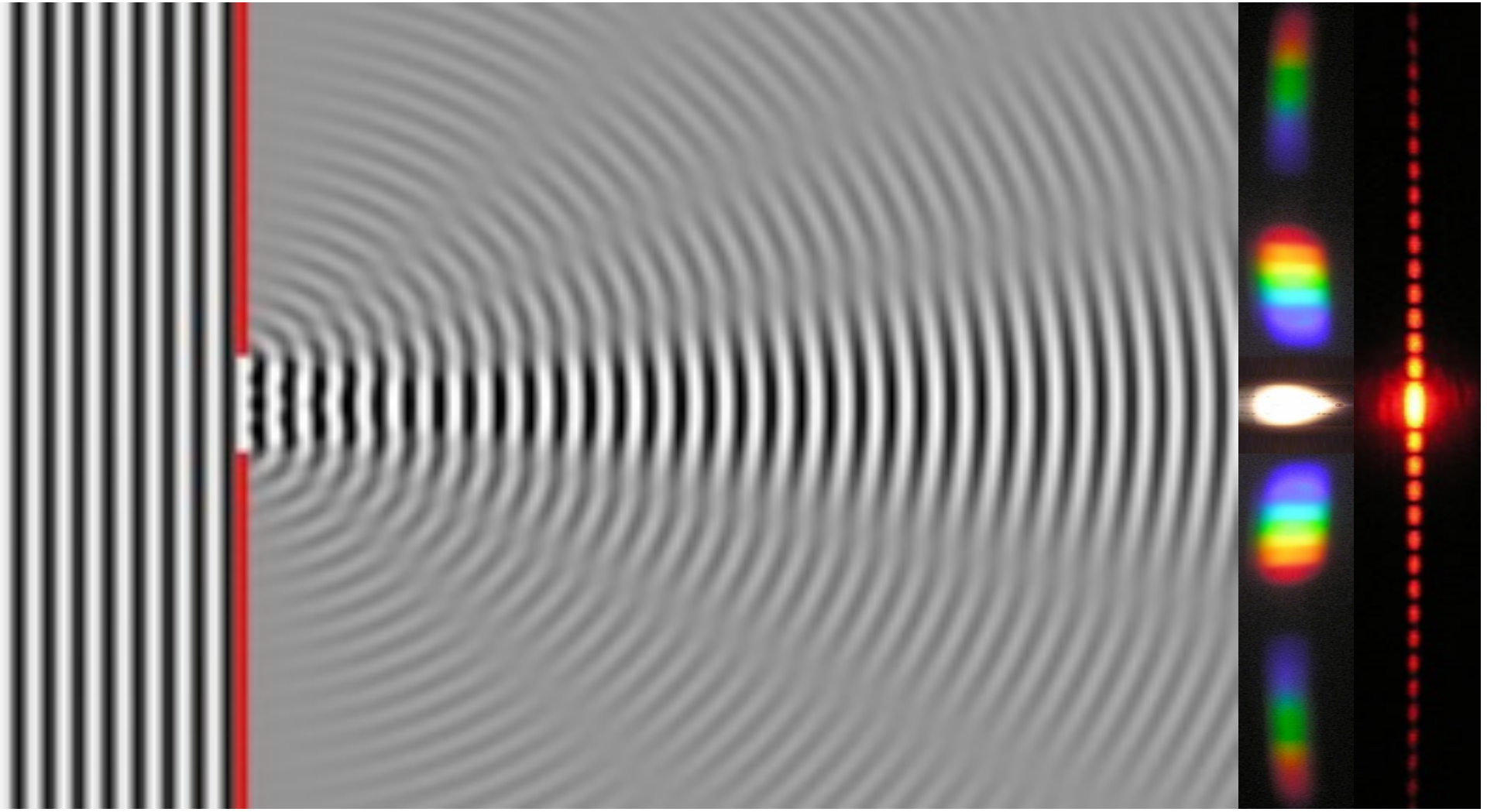




Newton's rings

Wave Behavior, Young (1801)

Single slit Experiment



Maxwell's Derivations (1864)



$$\rho = 0 \qquad J = 0$$

$$\nabla \cdot E = 0 \qquad \nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times B = \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

Wave equation

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\frac{\partial^2 B}{\partial y^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial x^2}$$

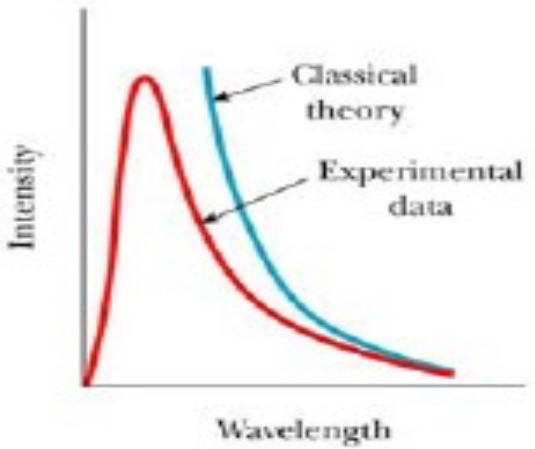
$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

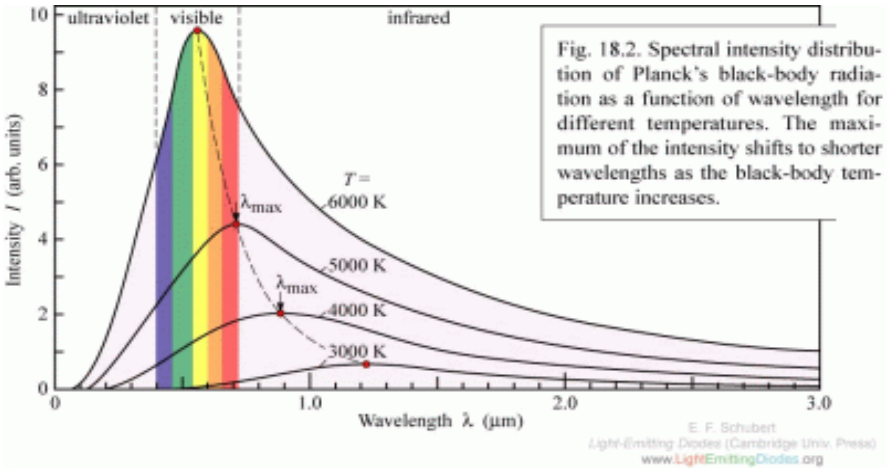
ε_0 = electric permittivity

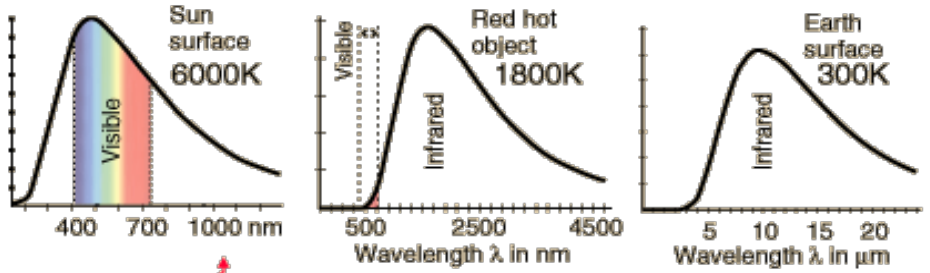
μ_0 = magnetic permeability

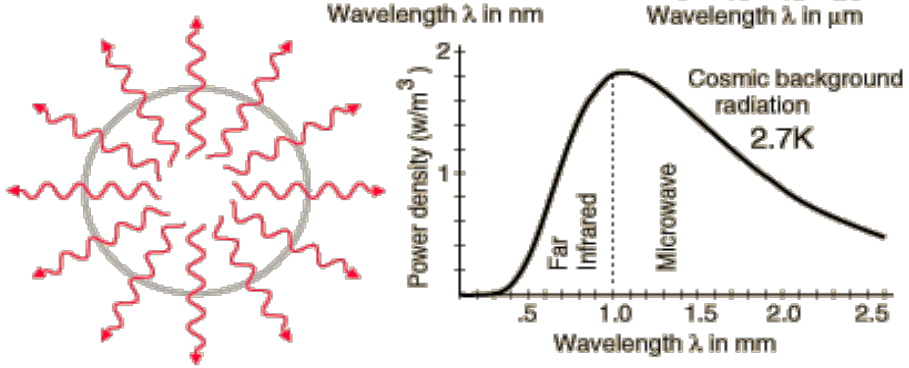
Plank's Findings (1900)









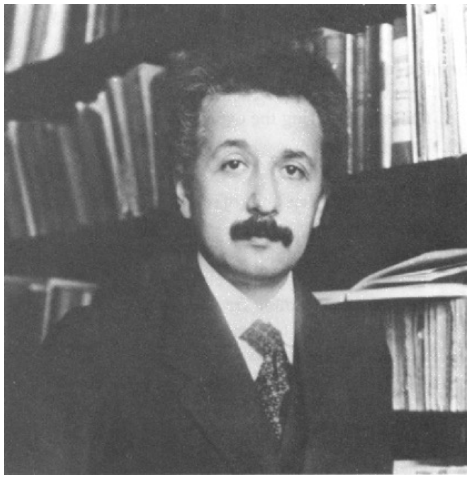


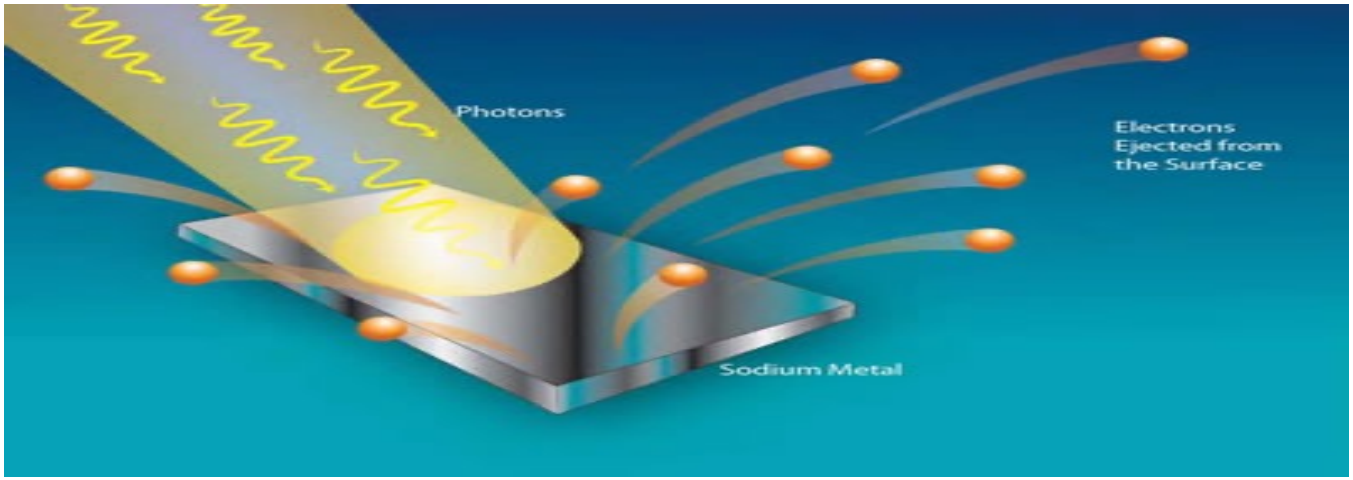
Blackbody radiation Experiment

$$I(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{kT}} - 1}$$

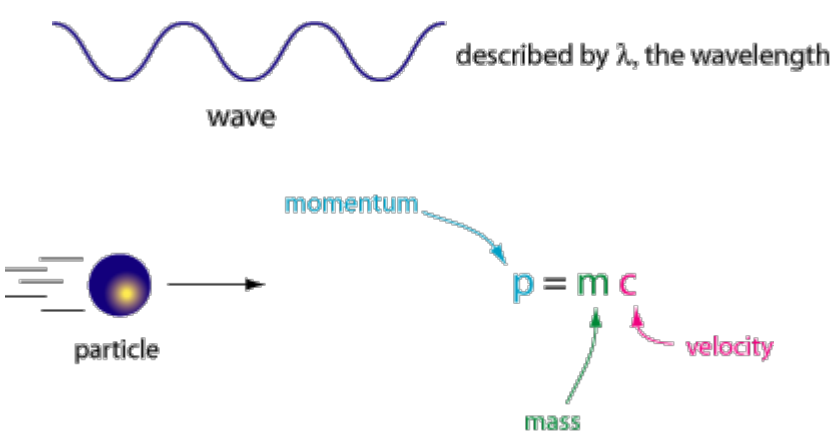
$$E = nhf$$

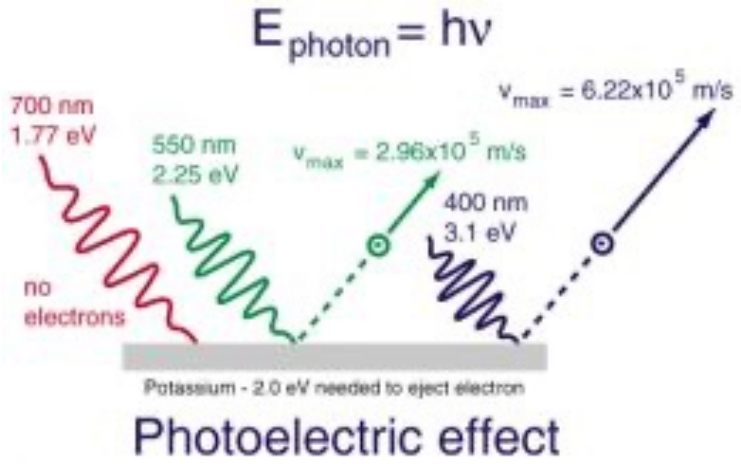
Einstein's Equations (1905)





The energy of the electrons ejected from the surface depends **not at all on the intensity of light** but only on the **wavelength**





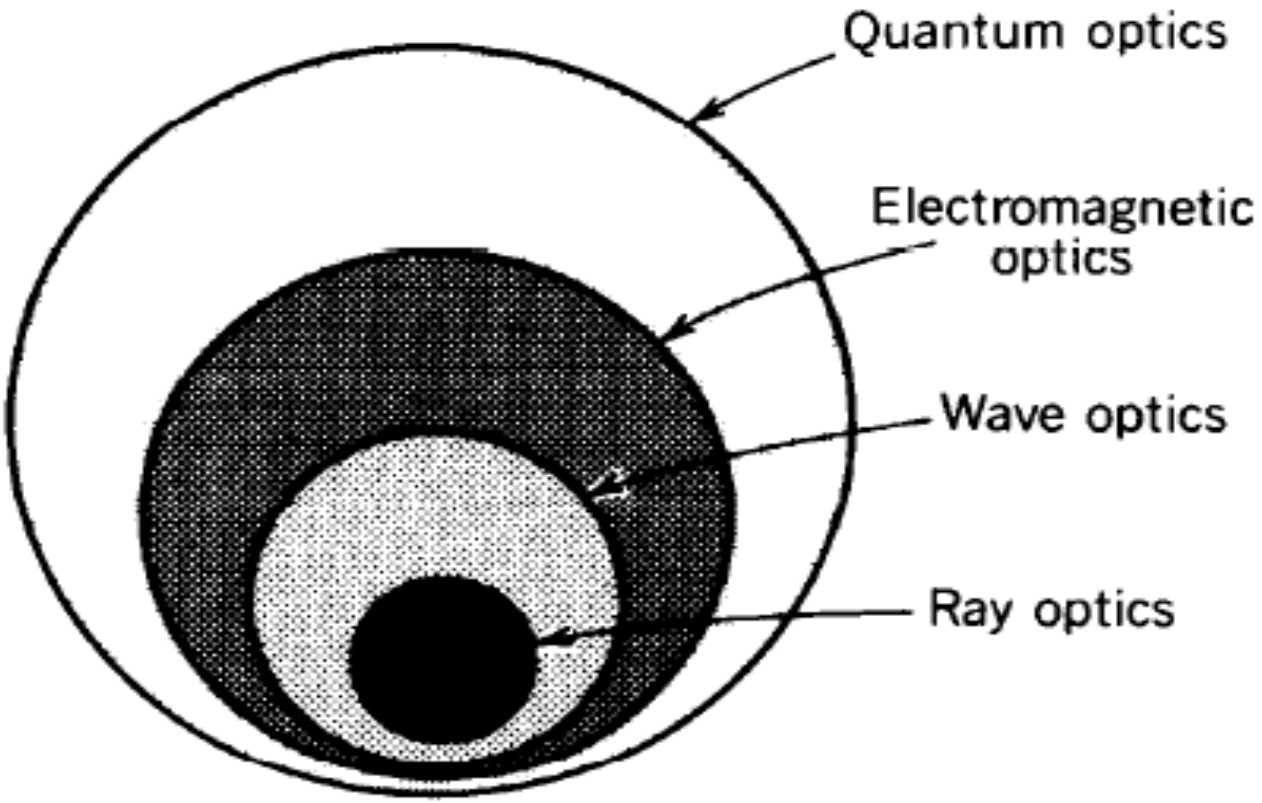
Light Mixed Behavior

Wave-particle duality

Property of EM radiation	Consistent with WAVE	PARTICLE
Reflection	Yes	Yes
Refraction	Yes	Yes
Interference	Yes	No
Diffraction	Yes	No
Polarization	Yes	No
Photoelectric effect	No	Yes

So, How to Deal with Light?

Size-Wavelength relationship



Ray Optics

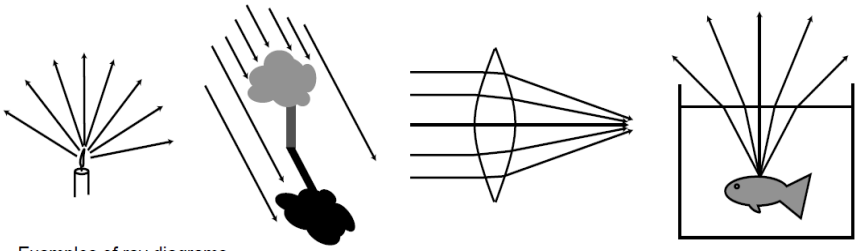


Ray Optics

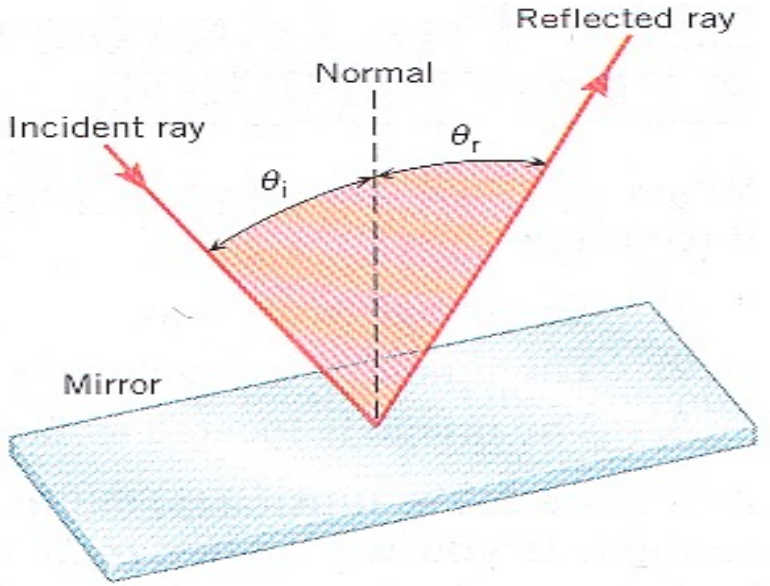
Objects and distances are much larger than optical wavelength $l >> \lambda$

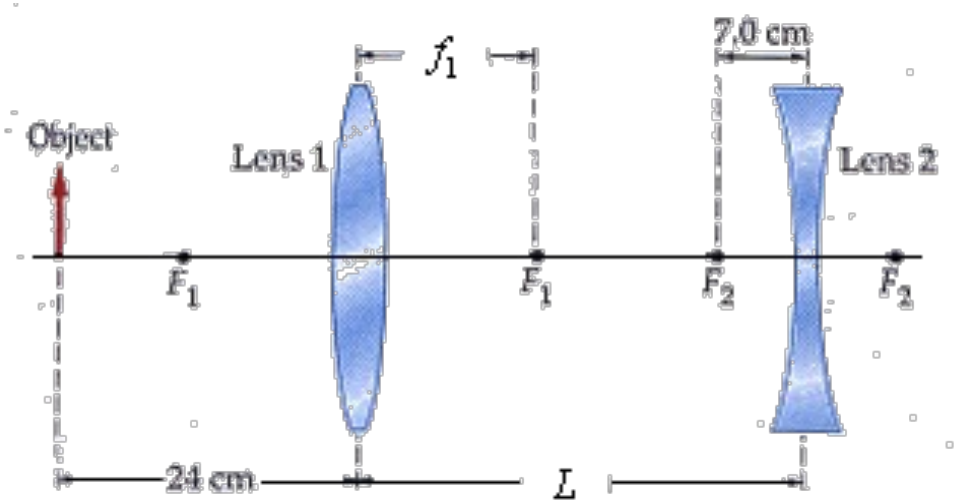
Material represented by

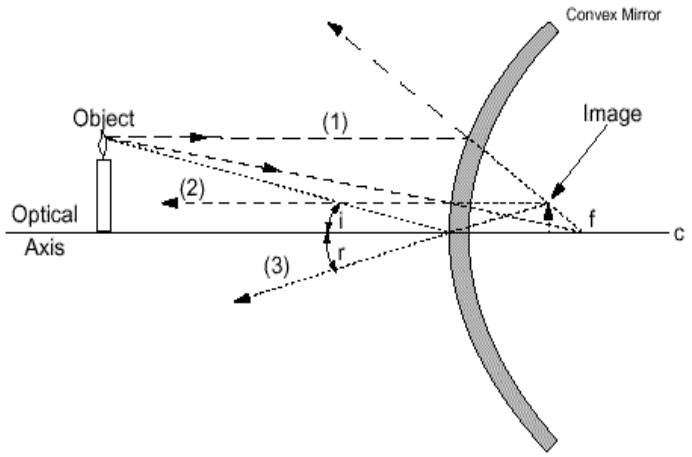
Refractive Index (n)



Examples of ray diagrams.





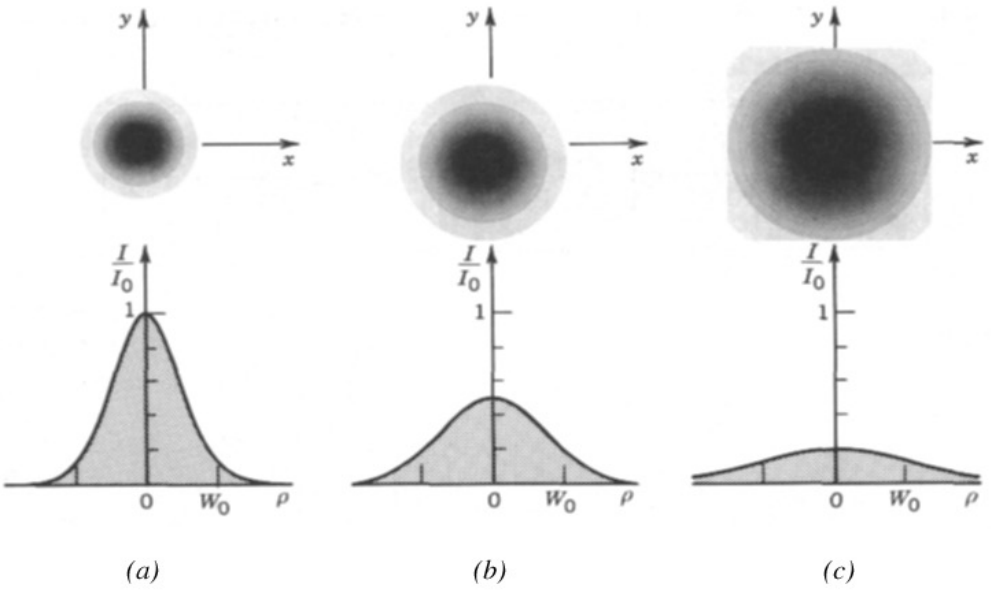


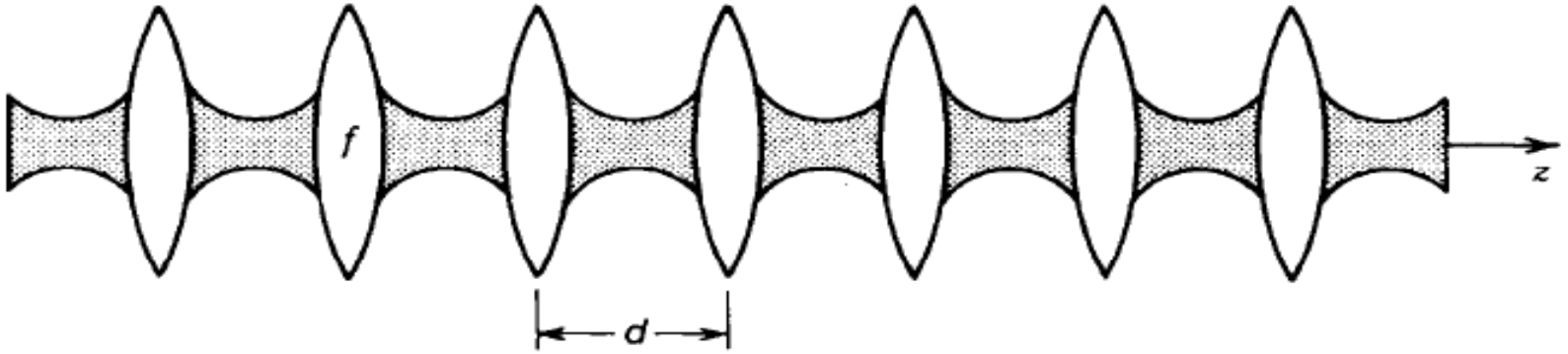
Beam Optics

Objects and distances are much larger than optical wavelength $l >> \lambda$

Material represented by **Refractive Index** (n)

Collective picture of ray bundles.





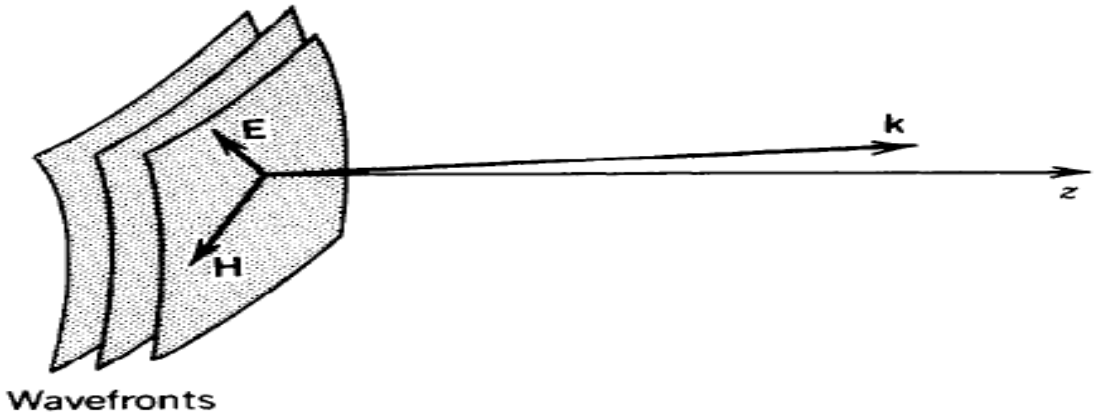
Wave Optics

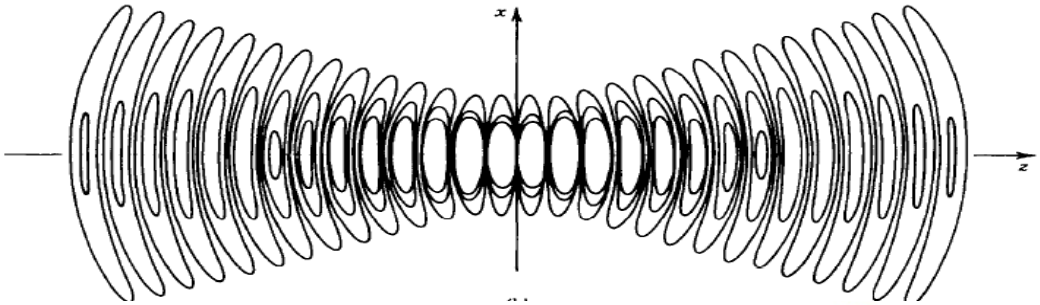
All length scales
except atomic scales

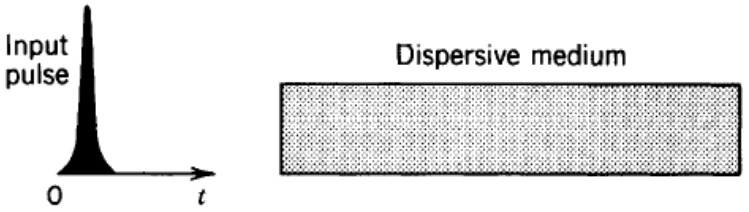
EM Field Theory
The Wave equation
Maxwell's equations
Light behaves wave-like

Macroscopic
Material properties!!

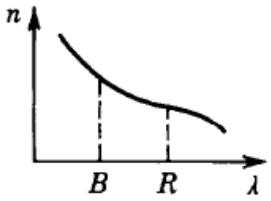
Complex Refractive Index

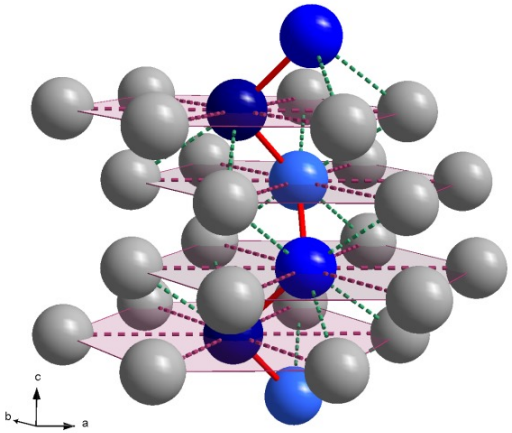










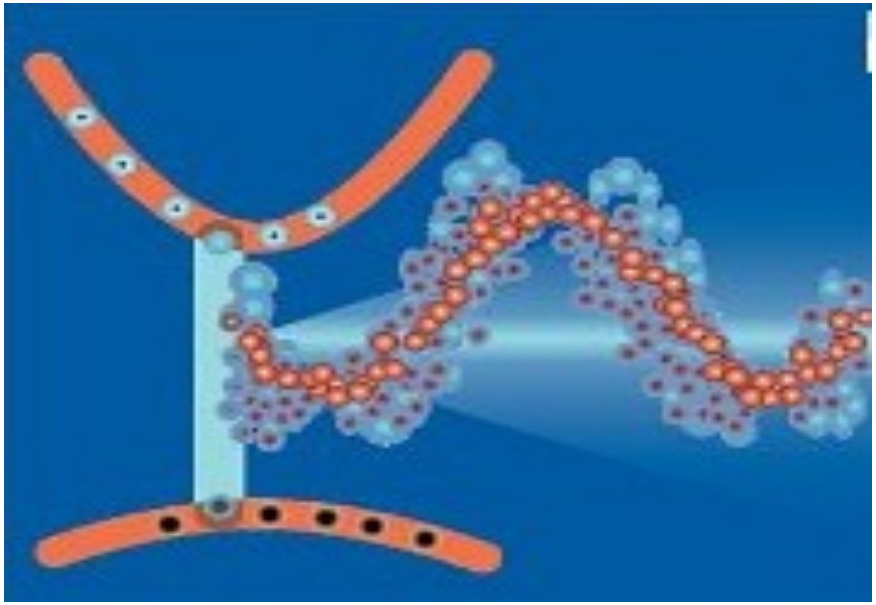


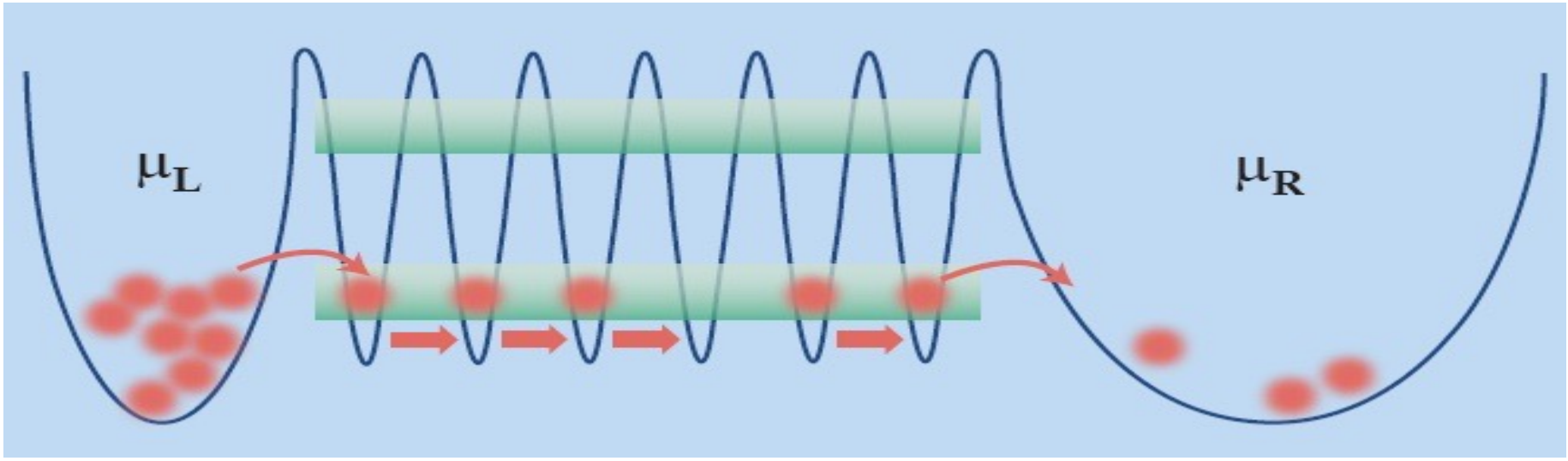
Quantum Optics

Atomic scales and energies

Material represented by
Energy Levels

Light behaves particle-like
Resonances, Absorption and emission





Q & A

Send me an email if you have any questions.